LESSON 3.2

Understand Rates and Unit Rates

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can describe a rate using its two different units and use a model to find a travel time, and I can find a unit rate to compare prices and decide the better buy.

WHY THIS MATTERS You're comparing ticket prices at different game booths in the arcade. Booth A charges $3 for 5 games, and Booth B charges $5 for 8 games. Which booth gives you a better deal? To find out, you need to figure out the cost per game — that's the unit rate!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Rates and Unit Rates *Spanish: Tasas y tasas unitarias*	A rate compares different units; a unit rate compares a quantity to 1 unit.	180 miles in 3 hours = 60 miles per 1 hour.
Rate *Spanish: Tasa*	A ratio comparing two amounts with different units, like miles per hour.	96 pages / 3 hours
Unit Rate *Spanish: Tasa unitaria*	A rate for just 1 of something, like cost for 1 item.	32 pages per 1 hour
Per *Spanish: Por*	For each one. Example: 5 dollars per book.	miles per hour, cost per pound
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	5 to 3 or 5:3
Unit price *Spanish: Precio unitario*	A unit rate that gives the cost of exactly 1 item, like $0.60 per game.	$3 for 5 games → $0.60 per 1 game

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A printer prints 90 pages in 6 minutes at a steady rate. How many pages in 7 minutes? How long for 45 pages?

1. Divide: 90 ÷ 6 = 15 pages per minute.
2. 7 minutes: 15 × 7 = 105 pages.
3. 45 pages: 45 ÷ 15 = 3 minutes.

Answer: 105 pages; 3 minutes

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A booth charges $4 for 8 games. What do we divide?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The rate is _____ for every _____ .
- I divided _____ by _____ to find the unit rate.
- The unit rate is _____ per _____ .
- To find the amount for _____ , I _____ .

Word bank rate | unit rate | per | each | divide | multiply | for every | speed | predict | constant | hour | pound

WATCH OUT!

A common mistake

A common mistake in Rates and Unit Rates is comparing total costs instead of unit rates: a student sees Pack X at $4.00 for 10 tokens and Pack Y at $6.00 for 16 tokens and picks Pack X just because $4.00 is the smaller number, without dividing to find the cost per token ($0.40 for Pack X vs. $0.375 for Pack Y

LESSON 3.2

Understand Rates and Unit Rates

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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WHY THIS MATTERS You're comparing ticket prices at different game booths in the arcade. Booth A charges $3 for 5 games, and Booth B charges $5 for 8 games. Which booth gives you a better deal? To find out, you need to figure out the cost per game — that's the unit rate!

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the unit rate for each arcade deal.

Deal	Total Cost	Number of Games	Unit Rate
Deal A	$6.00	12 games	$0.50 per game
Deal B	$7.50	10 games	
Deal C	$4.80	8 games	

HINT Look at Deal A's finished row: $6.00 for 12 games became $0.50 per game. Your rows work the same way.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A booth charges $4.50 for 9 games. What is the unit rate?

(A) $0.45 per game

(B) $0.50 per game

(C) $2.00 per game

(D) $4.50 per game

HINT You know the total cost ($4.50) and the total games (9). A unit rate tells the cost of just ONE game.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

(A) 6 miles per gallon

(B) 30 miles per gallon

(C) 36 miles per gallon

(D) 180 miles per gallon

HINT Notice the units in the question: miles PER gallon. That tells you miles go on top and gallons on the bottom.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

4 **Circle the letter of the best answer. Show how you know.**

A pack of 8 markers costs $6.00. What is the cost per marker?

(A) $0.60
(B) $0.75
(C) $0.80
(D) $1.33

HINT You have 8 markers costing $6.00 total. 'Cost per marker' means the price of ONE marker.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

5 **Circle the letter of the best answer. Show how you know.**

Which is a unit rate?

(A) 12 miles per 1 hour
(B) 24 miles in 2 hours
(C) 36 miles for 3 gallons
(D) 100 points in 5 games

HINT A unit rate always compares to exactly ONE of something — look at the second number in each choice.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read the problem** A booth charges $6.00 for 4 games. Find the unit rate (cost per game).
2. **Set up the division** $6.00 ÷ 4 = $1.50, but the student writes $1.05 per game.
3. **State the unit rate** The student says each game costs $1.05.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Check the place value: is the answer about one dollar and a half, or one dollar and five cents?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does finding the unit rate help you compare the booths?

SENTENCE STARTER The best deal is Booth ___ because its unit rate of ___ per game is the ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.2

Understand Rates and Unit Rates

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table to find unit rates and determine the best deal.

Deal	Quantity	Cost	Unit Rate
Tokens Pack A	20 tokens	$5.00	
Tokens Pack B	35 tokens	$7.00	
Tokens Pack C	50 tokens	$12.50	

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each situation by whether the unit rate is 'per item' or 'per time'.

Groups: Rate Per Item | Rate Per Time

_____ $3 per pound of apples

_____ 60 miles per hour

_____ $0.25 per token

_____ 15 pages per minute

_____ $1.50 per bottle

_____ 8 laps per hour

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Token packs: Pack A is 20 tokens for $5.00, Pack B is 35 tokens for $7.00, and Pack C is 50 tokens for $12.50. Which pack is the best deal per token?

(A) Pack B

(B) Pack A

(C) Pack C

(D) They are all the same

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?

(A) Runner B

(B) Runner A

(C) Same rate

(D) Cannot determine

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read the problem** Pack X is 10 tokens for $4.00. Pack Y is 16 tokens for $6.00. Which is the better buy (lower cost per token)?
2. **Compare the deals** Pack X costs $4.00 and Pack Y costs $6.00. The student says Pack X is the better buy because $4.00 is the lower total price.
3. **State the answer** The student chooses Pack X.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does finding the unit rate help you compare the booths?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.2

Understand Rates and Unit Rates

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Two water fountains fill bottles at different rates. Fountain A fills 3 bottles in 5 minutes. Fountain B fills 7 bottles in 10 minutes. Compare their rates.

Fountain	Bottles	Minutes	Bottles per Minute
Fountain A	3	5	
Fountain B	7	10	
Faster?			

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Rate Error

1. **Find the unit rate: 12 tickets for $8** 12 tickets for $8
2. **Set up the division** 12 ÷ 8 = 1.5
3. **Write the unit rate** The unit rate is $1.50 per ticket

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

An arcade sells tokens in three packs: 25 tokens for $5, 60 tokens for $10.80, and 100 tokens for $19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How does finding the unit rate help you compare the booths?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help